

House Price Prediction Using Machine Learning

Project Summary

The objective of this project was to develop a machine learning model capable of predicting house prices using property characteristics from a housing dataset containing 545 records and 13 features. The dataset included variables such as area, bedrooms, bathrooms, stories, parking availability, air conditioning, furnishing status, and other housing amenities.

The dataset was explored and assessed for quality before model development. No missing values or duplicate records were identified, reducing the need for extensive cleaning. Categorical variables were converted into numerical values using mapping techniques to make them suitable for machine learning algorithms. Histograms, scatter plots, and correlation heatmaps were created to understand the distribution of house prices and the relationships between different property features.

Analysis revealed that area, bathrooms, and air conditioning were among the strongest factors influencing house prices. Feature importance analysis showed that area alone contributed approximately 46.88% of the model's predictive power, followed by bathrooms at 15.30%. These findings indicate that larger homes with more bathrooms and desirable amenities generally command higher market values. An interesting observation was that some amenities expected to influence prices had a relatively limited impact when compared with property size and bathroom count.

Two machine learning models were developed and evaluated: Linear Regression and Random Forest Regression. The Linear Regression model achieved an MAE of ₹979,680, an RMSE of ₹1,331,071, and an R^2 score of 0.6495, while the Random Forest model achieved slightly lower performance. The results indicate that Linear Regression was able to explain approximately 65% of the variation in house prices. Although the model provides a useful estimate of property value, a portion of price variation is still influenced by factors not included in the dataset.

From a business perspective, the findings suggest that real estate organizations should place greater emphasis on property area, bathroom count, and key amenities when estimating market value. The model can serve as a practical decision-support tool for identifying potentially undervalued or overpriced properties and supporting more consistent, data-driven pricing strategies.

Conclusion

This project successfully demonstrated the use of machine learning techniques for house price prediction. After comparing both models, Linear Regression achieved better performance and was selected as the final model. The results highlight the importance of property size, bathrooms, and amenities in determining house prices and demonstrate how predictive analytics can support real estate decision-making.

